TRAN VUONG HOANG

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OVERVIEW

I am a research resident at Qualcomm AI Research and an AI Researcher at DreamAI Team. I earned my Bachelor's degree in Computer Science from Hanoi University of Science and Technology. My work is in Machine Learning and NLP, currently focused on Knowledge Distillation, Reinforcement Learning, Agentic AI, and Human–AI Alignment. I aim to grow as a researcher who works internationally and turns new methods into systems that hold up in practice.

EDUCATION

Hanoi University of Science and Technology Sep 2022 – 2026
School of Information and Communications Technology *Hanoi, Vietnam*
Computer Science (Talented Bachelor Program)

- **GPA:** 3.71 / 4.0

Ha Long High School for Gifted Students Aug 2019 – Jun 2022
Math honors class *Quang Ninh, Vietnam*

- **Grade:** 9.1 / 10

EXPERIENCES

Qualcomm Mar 2026 – Present

- ***AI Resident:*** Selected for a prestigious research program to advance AI in Vietnam, focusing on solving real-world challenges and developing impactful research alongside top industry professionals.

DreamAI Team Nov 2025 – Present

- ***AI Researcher:*** Conducting research on Knowledge Distillation, Human–AI Alignment, and reasoning in LLMs.

ZenAI Mar 2025 – Feb 2026

- ***AI Intern:*** Developing and fine-tuning Large Language Models (LLMs), building autonomous agents, and optimizing retrieval and text generation pipelines for AI solutions.

Data Science Laboratory (DSLab), HUST Jan 2024 – Nov 2025

- ***Research Student:*** Conducting research under Dr. Ngo Van Linh, focusing on Natural Language Processing, Knowledge Distillation, and Optimal Transport.

PUBLICATIONS

[1] **A Survey on Knowledge Distillation of Vision-Language Models** (First author) EMNLP 2026
[2] **f-Divergence Self-Play for Tabular Anomaly Detection via Large Language Models** (First author) ICML 2026
[3] **MCW-KD: Multi-Cost Wasserstein Knowledge Distillation for Large Language Models** (First author) AAAI 2026
[4] **Token-Level Self-Play with Importance-Aware Guidance for Large Language Models** (Author) NeurIPS 2025
[5] **Multi-Surrogate-Objective Optimization for Neural Topic Models** (Co-first author) EMNLP 2025
[6] **HiCOT: Improving Neural Topic Models via Optimal Transport and Contrastive Learning** (First author) ACL 2025
[7] **Sharpness-Aware Minimization for Topic Models with High-Quality Document Representations** (Co-first author) NAACL 2025
[8] **A Framework for Neural Topic Modeling using Hierarchical Clustering and Contrastive Learning with Optimal Transport** (Co-first author) Neurocomputing (Q1)
[9] **OrthoSAM: Orthogonal Perturbations for Sharpness-Aware Minimization** (First author) Submitted to NeurIPS 2026

PROJECTS

A Survey on Knowledge Distillation of Vision-Language Models (Link)

- Surveyed knowledge distillation for vision-language models with a taxonomy along knowledge type, teacher accessibility, and structural compatibility, covering datasets, evaluation protocols, and open challenges.

DiSPaT: f-Divergence Self-Play for Tabular Anomaly Detection via Large Language Models (Link)

- Cast unsupervised tabular anomaly detection as f-divergence minimization and refined an LLM detector via self-play on real and generated samples, outperforming AnoLLM and strong baselines.

MCW-KD: Multi-Cost Wasserstein Knowledge Distillation for Large Language Models (Link)

- Built a multi-cost Wasserstein distillation framework that aligns hidden states and output distributions across heterogeneous LLMs, improving ROUGE-L and LLM-based evaluations.

HiCOT: Improving Neural Topic Models via Optimal Transport and Contrastive Learning (Link)

- Combined Optimal Transport and contrastive learning to align topic-word distributions and sharpen topic semantics, improving coherence, diversity, and document representations over strong baselines.

HiCOLT: A Framework for Neural Topic Modeling using Hierarchical Clustering and Contrastive Learning with Optimal Transport (Link)

- Aligned PLM document embeddings with topic distributions via Optimal Transport and a hierarchical contrastive module with triple-level objectives to refine topics and improve coherence.

HONORS AND AWARDS

Odon Vallet Scholarship Aug 2026
Association Rencontres du Vietnam

The Best of MB Chasing Scholarship Dec 2024
MB Bank

Vietnam National Mathematical Olympiad (VMO) 2020 – 2022
Participant (2020–2021 and 2021–2022)

- Represented the province at the national olympiad for two consecutive years.

Hong Kong International Mathematical Olympiad (HKIMO) 2022
Gold Medal (National Round)

- Won Gold in the Vietnam National Round and qualified for the International Final.

Provincial Mathematical Olympiad 2020 – 2022
First Prize (2021–2022), Third Prize (2020–2021)

- Annual competitive examination to select the most talented mathematics students in the province.

ACTIVITIES

SOICT Hackathon 2024 – Traffic Vehicle Detection (NAVER) track 2024
Participant — built Computer Vision models for traffic vehicle detection under complex conditions.

Student Creative Ideas Challenge (SCIC) 2023
Finalist — selected for the final round among student technical projects at HUST.

"Student of 5 Merits" Award 2023
City-Level Distinction — recognized for ethics, academics, fitness, volunteering, and integration.

SKILLS

Technical Skills: Deep Learning, Natural Language Processing, Reinforcement Learning, Knowledge Distillation, Optimal Transport, Agent, Sharpness-Aware Minimization, Direct Preference Optimization, Topic Models.
Programming Languages: Python, C/C++, SQL.
Tools/Others: PyTorch, Git, MySQL, Docker, Conda, LaTeX, Wandb, FastAPI, vLLM, Runpod.